

Interoperable Immunization Data Initiative

Overview of Phase 1 Proof of Concept (PoC)

April 2026

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Objectives

- To provide business context and changes that have occurred since the Interoperable Immunization Data Initiative (IIDI) Phase 1 Proof of Concept (PoC) was released, with an emphasis on the shift to conceptualizing the Pan-Canadian Health Data Highway.
- To articulate an overview of the Phase 1 PoC that was completed in March 2025.
 - NOTE: As the focus of Phase 1 is to demonstrate technical feasibility in a virtualized environment, all user interfaces that are depicted in this demo are simply for **illustrative purposes** and are not meant to be used by PHAC or PT partners.

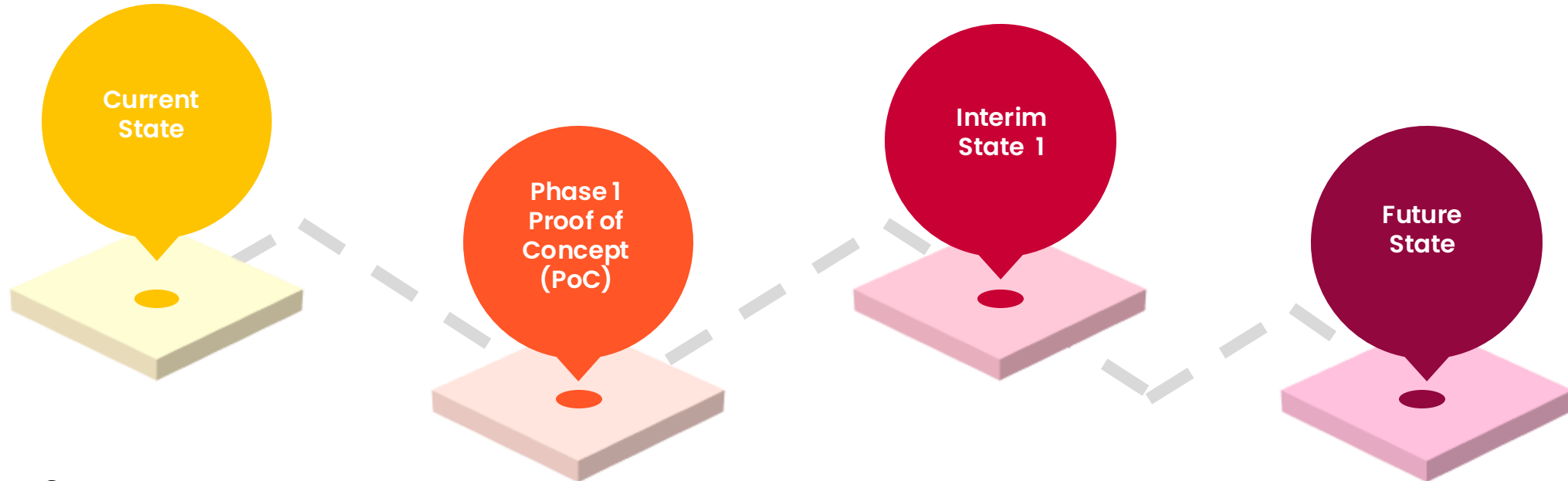
Context

Phase 1 PoC and Conceptualizing the Way Forward

- ❑ **Phase 1 Proof of Concept (PoC).** Phase 1 PoC simulated data sharing amongst PTs and with PHAC with BC and ON as pilot sites. This was delivered March 2025.
- ❑ **Architecting the Pan-Canadian Health Data Highway.** IIDI has shifted away from continued PoC testing, and instead undertaking a conceptual exercise for the required inputs to scale the PoC to a future interim state, which is referred to as the Pan-Canadian Health Data Highway or Interim State 1.

Pan-Canadian Health Data Highway - Architecting for the Future

Instead of continuing with the testing of the PoC, IIDI has shifted its efforts to define a high-level architecture for Interim State 1.



Current State

- Immunization data is siloed to varying extents across PTs.[^]
- HMM (Health Minister Meeting) commitment to explore Federated Data Architecture for immunization coverage data

Phase 1 PoC (March 2025)

- PoC focused on baseline technical feasibility featuring simulated data access amongst BC, ON and PHAC for childhood immunizations (MMR)

Interim State 1 (2-5+ years)

- Pan-Canadian Health Data Highway using federated data to connect immunization data across jurisdictions, while maintaining P/T data sovereignty

Future State (5-10+years)

- Data highway expanded to public health data beyond immunization, and with other GC clients

[^] PTs manage immunization data differently thus immunization data interoperability varies across jurisdictions.

The Shift: Two New Streams of Work

Building on the results and lessons learned of the PoC, PHDSC recommendation and PTs feedback
=> Opportunity to address the barriers to data interoperability by extending our focus from the technology pillar to governance and data stewardship enablement



Data Stewardship and Governance Enablement (Policies & Governance, Standards)

Data-Flow Gap Analysis

Leveraging the CIHI Data Stewardship Framework and the technical Proof of Concept, identify interoperability enablers and barriers and advance recommendations



Technical Assessment Readiness (Technology)

PoC Scalability Assessment

Undertake assessments to determine if the federated data Proof of Concept can scale to Interim State 1 and what inputs would be required (*Pan-Canadian federated data exchange framework*)^

PT Technical Readiness Assessment

Conduct PT/Federal Technical Readiness Assessments to determine Canada's foundational readiness to adopt federated data in immunization

Phase 1 PoC

Phase 1 PoC Evaluation Framework/Requirements Traceability Overview

WHY

To demonstrate the feasibility and opportunities of federated data as a technological solution to automate data processes and enhance workflow efficiencies in support of the Joint FPT Action Plan priority to increase access to quality immunization data for Canadians, healthcare providers and local public health.

Examples of expected outcomes for the post-PoC operationalized state



Enhanced efficiency of PT/PHAC data workflows



Streamlined data access and collation



Controlled access while maintaining data sovereignty

WHAT

High-Level Functionalities

- 1 Federated data technology is designed and adapted based on FPT requirements identified by the Sub-Working Group
- 2 Federated data technology functions meet the technical requirements
- 3 Use of synthetic data closely resembles data currently captured in PT immunization registries aligned with current standards
- 4 Immunization registries (current state) for both BC and ON are emulated

Functional Requirements/User Stories

- 🔍 High-level functionality statements are broken down into **functional requirements/user stories** for the two user journeys.
- 🔍 This represents the **"WHAT"** that is being built for the Phase 1 PoC and is socialized with all stakeholders to establish scope.

HOW

Implementation Approach

📄 Solution architecture (the **'HOW'**) is created and designed to meet the scope represented by the functional requirements /user stories for the two user journeys.

Validation

Acceptance Criteria & Testing

- ⚙️ Acceptance Criteria and testing scenarios are executed for functional requirements/user stories to ensure completeness and successful fulfillment
- ⚙️ Ultimately, validation is to ensure the **"WHY"** has been met



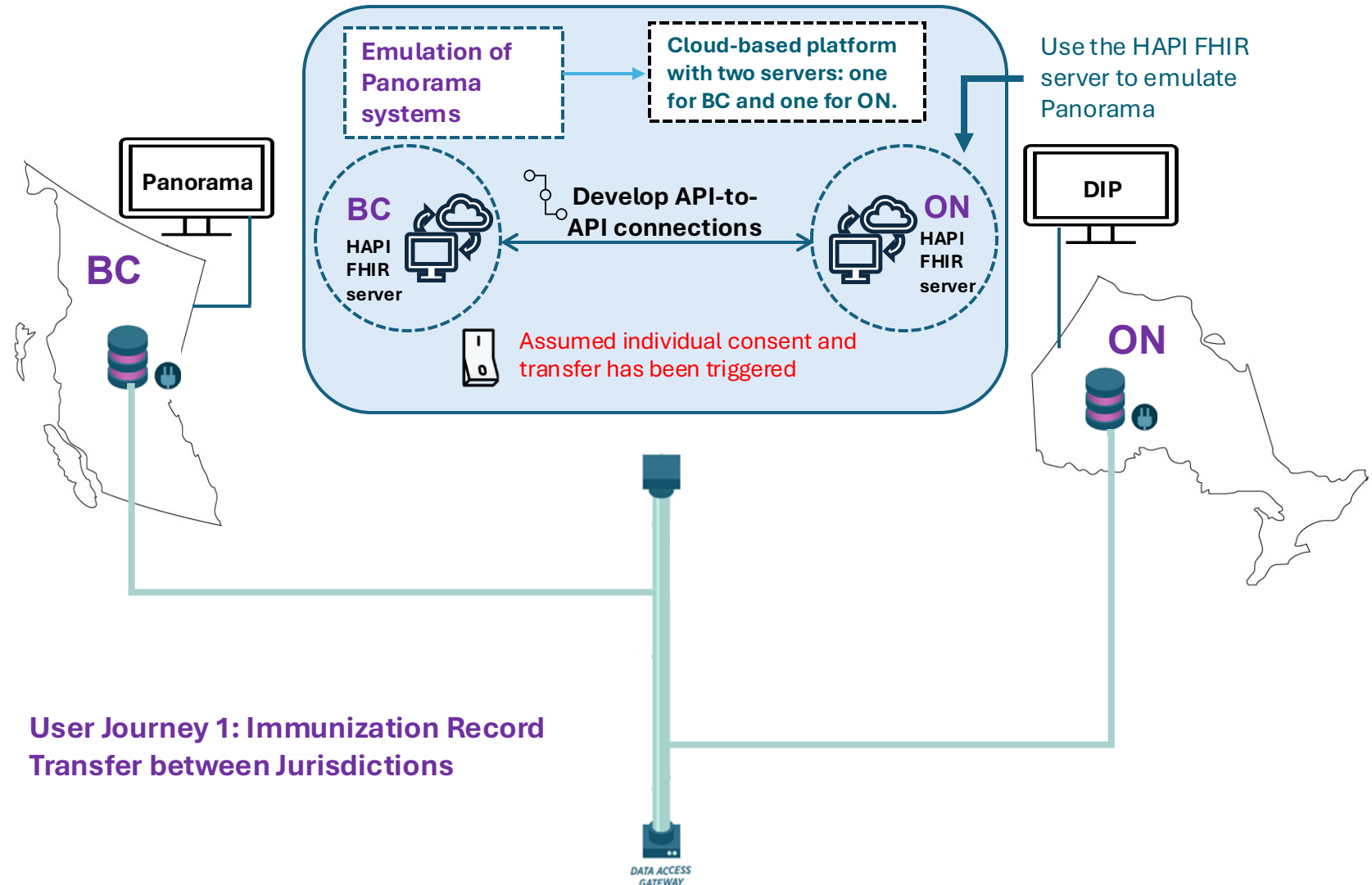
Phase 1 (March 31, 2025) Technical Scope

User Journey 1: Seamless Immunization Record Transfer Between Jurisdictions

A resident moves from one PT to another, provides consent and triggers transfer.

Assumptions (out-of-scope for Phase 1):

- Consent management is already in place
- Policies, legislations, data sharing agreement & data governance is in place
- Push-based model where the originating province (data owner) initiates the transfer only after an out-of-band consent/transfer authorization process is completed.





Phase 1 (March 31, 2025) Technical Scope

User Journey 1: Seamless Immunization Record Transfer Between Jurisdictions

In-Scope
For Phase 1



PHAC will create two HAPI FHIR servers within its cloud platform to emulate BC & ON's Panorama.



PHAC will generate synthetic data using agreed upon data elements from BC & ON's Panorama environment.



Simulate the automatic secure transfer of immunization record between BC & ON's Panorama using the federated data platform.

Out-of-Scope
For Phase 1



UI for individuals (optional), user identification, and consent management mechanisms.



Policies and legislations.



For this phase, it will assume individual has provided consent and a trigger has been made to initiate the immunization record transfer.



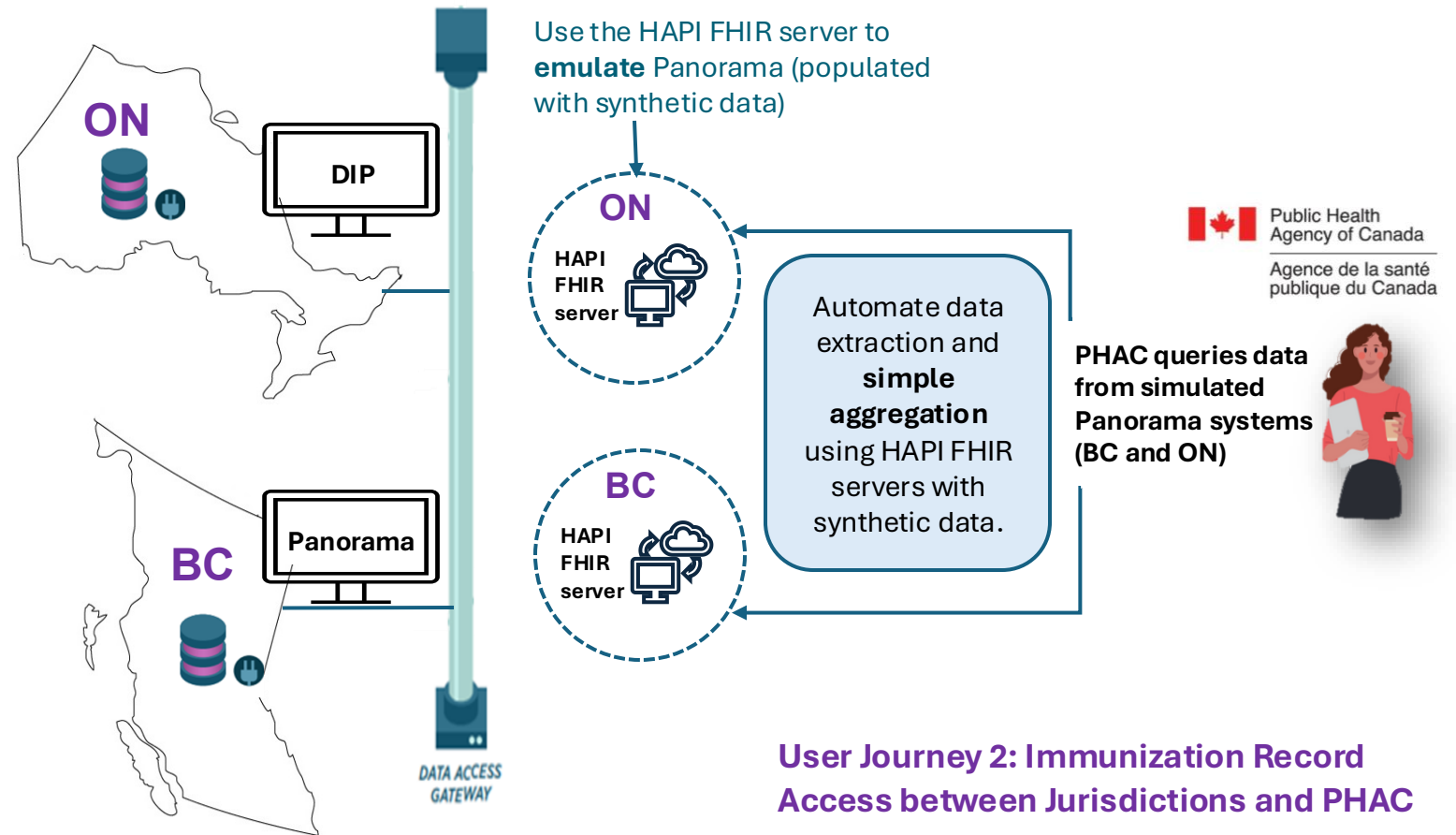
Phase 1 (March 31, 2025) Technical Scope

User Journey 2: Seamless Immunization Data Access Between Jurisdiction(s) and PHAC

A query has been made by PHAC to retrieve data from the simulated Panorama systems (BC and ON).

Assumptions (out-of-scope for Phase 1):

- Automatic data extraction is based on respective PT's immunization logic & guidelines
- Aggregation is based on PT & PHAC's standard
- Governance for data access control is in place
- Data reporting standards are in place



User Journey 2: Immunization Record Access between Jurisdictions and PHAC



Phase 1 (March 31, 2025) Technical Scope

User Journey 2: Seamless Immunization Data Access Between Jurisdiction(s) and PHAC

In-Scope for Phase 1



A query has been made by PHAC to retrieve data from the simulated Panorama systems (BC and ON).



Simulate the automatic extraction and simple data aggregation from BC & ON's Panorama (simulated through HAPI FHIR servers populated with synthetic data).



Aggregated data can be queried by PHAC through the federated data platform.

Out-of-Scope For Phase 1



Detailed inclusion/exclusion criteria for data extraction and aggregation.



Actual UI.



Connection to analytical software (e.g., SAS, R) for PHAC's epidemiologists.

IIDI Phase 1 PoC – Data Flow Overview

UJ1: PT-to-PT Transfer Data Flow (Ontario → BC)

Transfer Initiation: Ontario retrieves immunization records from the HAPI FHIR Server for a patient moving to BC.

Job Queued in Redis: The FHIR JSON payload is queued for asynchronous processing, ensuring fault tolerance.

Secure Data Transfer: Records are transmitted securely using mTLS encryption.

Transfer Processing: BC verifies API authentication and validates the payload against HL7 FHIR standards.

Data Storage in BC: Validated records are ingested into BC's HAPI FHIR Server, ensuring integrity and compliance.

UJ2: PT-to-PHAC Reporting (Ontario & BC → PHAC)

Data Aggregation: Each PT's Aggregator Service queries the HAPI FHIR Server for immunization data.

FHIR Validation & De-Identification: Data is de-identified and validated to ensure compliance. No PHI is transferred.

Secure Data Transfer: Data is sent to PHAC's Federator Service using mTLS encryption.

Data Standardization: PHAC's Federator Service normalizes records for consistency across PTs.

Visualization & Insights: Data is analyzed and displayed on the R-Shiny Dashboard to track trends and coverage.

IIDI Phase 1 PoC – Demo Portal View

1. **Demo Interface (POC):** Temporary web UI built to showcase and test ITI components under IIDI.
2. **Jurisdictions:** Simulated PT (Provincial/Territorial) and Federal environments for end-to-end testing.
3. **Patient Browser:** Simple viewer to browse synthetic patient immunization records from the FHIR registry.
4. **HAPI FHIR Server:** Simulates a provincial immunization registry using the FHIR standard.
5. **Dashboard:** Displays immunization records received from provinces via the Federator in R-Shiny Dashboard.
6. **Federator:** Connects to PT aggregators, pulls standardized data, and forwards to the dashboard.
7. **Transfer Dashboard:** Simulates PT-to-PT sharing of immunization records for interoperability testing.

Interoperable Immunization Data Initiative (IIDi) / Initiative de Données d'Immunisation Interopérables (IDII)

Title

Disclaimer (EN): This web interface is temporary and serves as a **Proof of Concept (POC)** to facilitate current technical development and testing.

Avis (FR) : Cette interface web est temporaire et sert de **Preuve de Concept (POC)** pour faciliter les développements techniques actuels et les tests.

	1	2	3	4	5	6
Federal					Dashboard	Federator
British Columbia (BC)		Patient Browser	HAPI FHIR Server	Patient Transfer Dashboard		
Ontario (ON)		Patient Browser	HAPI FHIR Server	Patient Transfer Dashboard		

Demo link: <https://demo-portal.iidi.beta.phac.gc.ca/>

Synthetic Patient Data Generation

- The synthetic data mimics real-world immunization records from provincial registries while ensuring FHIR-compliant JSON format for interoperability.
- Differences between BC and ON data models are accounted for, including variations in fields such as allergy tracking.

Provincial Differences in Data Generation

British Columbia (BC) Specific Fields

- Allergy information is included using FHIR `AllergyIntolerance` resource.
- Uses SNOMED CT-coded allergy types, severity levels, and reaction dates.

Ontario (ON) Specific Fields

- Ontario does not track allergy information in immunization records.
- The script skips allergy generation for ON patients.

- Synthetic data used aligns with ON and BC data schemas and with the Immunization Registry Functional Standards (IRFS)

FHIR Patient Resource Structure

A FHIR Patient Resource consists of core fields and extensions, ensuring consistency across jurisdictions.

Field	FHIR Path	Example Value	Logic
Patient ID	Patient.id	"patient-001"	Unique identifier.
Name	Patient.name	{ "family": "Singh", "given": ["Simar"] }	Randomized names using Faker.
Gender	Patient.gender	"male", "female", "other"	Random selection.
Birth Date	Patient.birthDate	"2012-06-15"	Randomized within given age range.
Address	Patient.address	{ "city": "Toronto", "state": "ON" }	Province-specific logic.

Immunization Data Generation (FHIR Standard)

Field	FHIR Path	Example Value	Logic
Vaccine Type	Immunization.vaccineCode	"MMR", "Influenza"	Random selection.
Manufacturer	Immunization.manufacturer	"Pfizer", "Moderna"	Random manufacturer assignment.

Summary of Key Features

- **FHIR-Compliant:** Structured to align with HL7 FHIR.
- **Synthetic but Realistic:** Uses Faker for realistic patient data.
- **Handles Provincial Variations:** BC and ON have different data models.
- **Includes Adverse Reactions & Exemptions:** Adds realism for testing.

IIDI Phase 1 PoC – Demo Transfer View

- 1. **Transfer UI Overview:** This screen simulates a PT-to-PT record transfer — from Ontario to another province.
- 2. **Patient ID Input:** Enter the ID of a synthetic patient in Ontario's system whose record will be transferred.
- 3. **Receiving PT Selector:** Choose the destination province (BC) to send the immunization record.
- 4. **Initiate Transfer:** Triggers a simulated, secure transfer using FHIR standards and mTLS encryption.
- 5. **Transfer Requests Table:** Displays submitted jobs with status, timestamps, and confirmation from the receiving PT.
- 6. **Refresh Button:** Updates transfer request statuses to reflect real-time completion or errors.
- 7. **Result Column:** Shows confirmation from the receiving PT, including the new Patient ID assigned on their end.

PT-to-PT MMR Immunization Record Transfer: Ontario

Initiate New Transfer

Patient ID 2
ID of a patient in Ontario's system

Receiving PT
Province or Territory to transfer records to

British Columbia

Initiate transfer

 4

Transfer Requests

 5

Refresh

 6

Patient ID	Receiving PT	Transfer Job ID	Initialized At	Transfer Status	Completed At	Result
3052	British Columbia	1	2025-03-25, 4:12:34 p.m. EDT	Success	2025-03-25, 4:12:42 p.m. EDT	Patient ID in receiving system: 7552

Summary of Ontario's historical and ongoing patient transfers.

Last refreshed: 2025-03-25, 8:19:14 p.m. EDT

1

3

4

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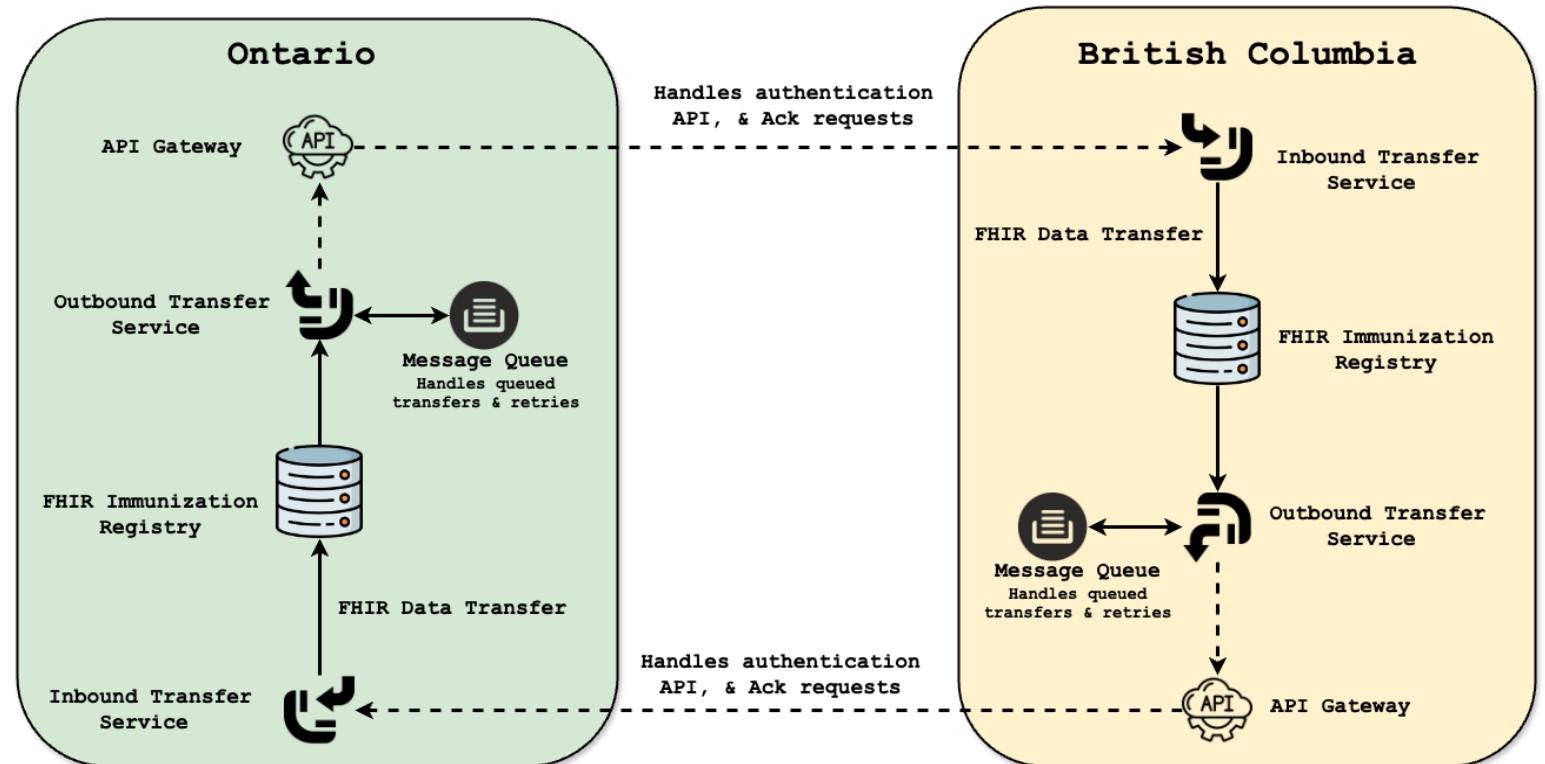
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Overview of User Journey 1: PT to PT Immunization Data Transfer

The diagram on the right illustrates how immunization records transfer between provinces (Ontario to BC example) when a patient relocates or seeks care elsewhere.

Key Components & Their Roles:

- **FHIR Immunization Registry:** Stores structured records, ensuring FHIR-compliant standardization.
- **API Gateway:** Manages authentication & secure API requests.
- **Outbound Transfer Service:** Extracts records & pushes transfers through a secure workflow.
- **Message Queue (Redis):** Manages queued transfers & retries for reliability.
- **Inbound Transfer Service:** Validates & writes received records into the registry.



Note: The PT-to-PT immunization data transfer in this demonstration uses synthetic data with unique patient IDs for consistency. This PoC does not include patient identification, record duplication checks, data quality assurance, or clinical validation. The sole focus is on demonstrating secure data transfer between British Columbia and Ontario, with real-world governance and validation processes out of scope.

Overview of User Journey 2: PT to PHAC Data Reporting

This diagram on the right shows how MMR immunization data flows from PT systems (Ontario & BC) to PHAC, ensuring local control of records while PHAC receives aggregated insights for surveillance and reporting.

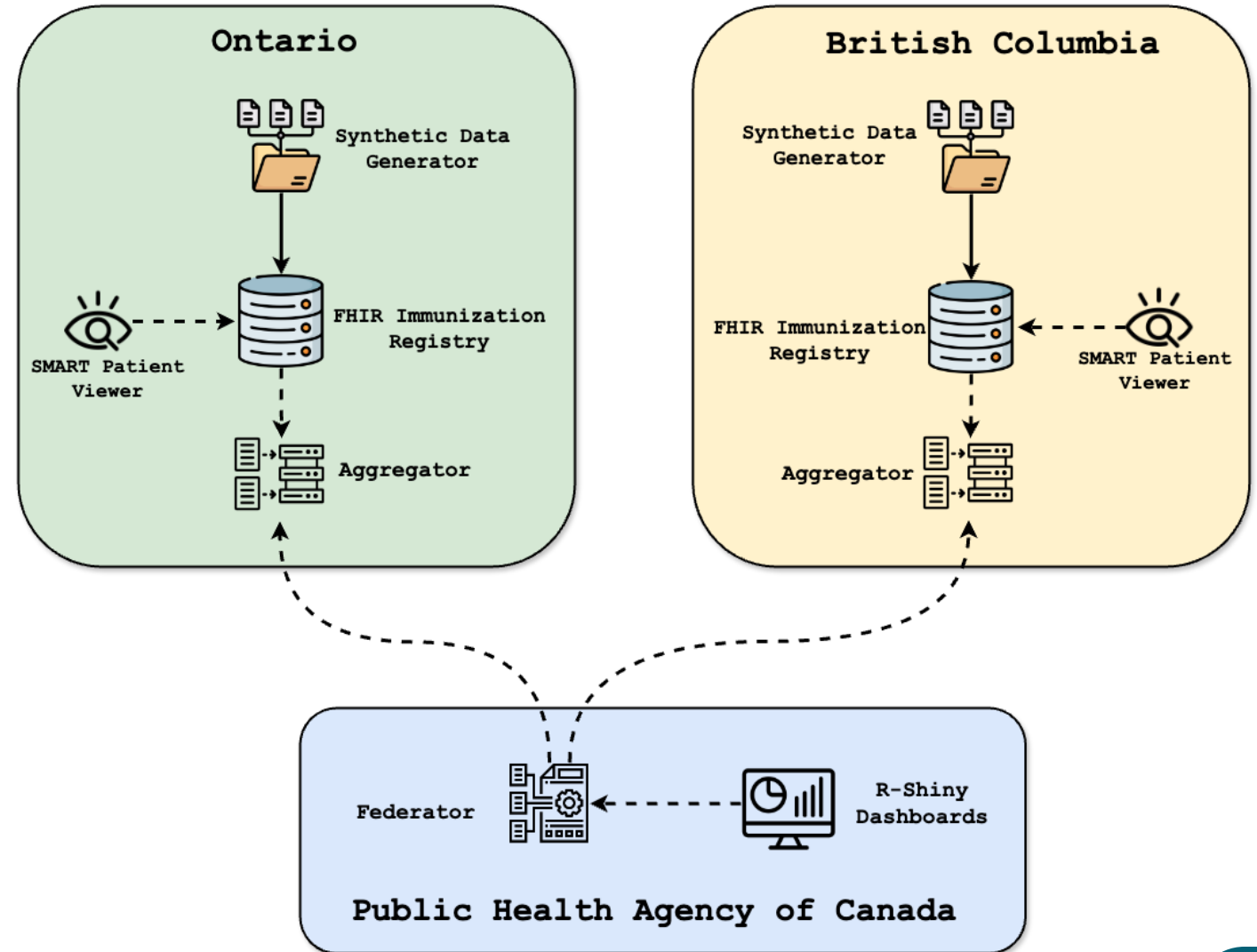
Key Components & Their Roles:

Each PT controls its own MMR records while enabling secure, federated data exchange.

- **Synthetic Data Generator:** Generates test MMR records for validation & PoC.
- **FHIR Immunization Registry:** Stores MMR records in a structured, FHIR-compliant format.
- **SMART Patient Viewer:** Allows PTs to securely view MMR records via FHIR APIs.
- **Aggregator:** Collects & processes MMR data from EMRs and prepares aggregated data for secure transmission to PHAC.

PHAC only receives aggregated MMR data for analysis—no individual patient records.

- **Federator:** Receives aggregated MMR data from PTs and ensures secure, privacy-compliant data transmission.
- **R-Shiny Dashboards:** Visualizes MMR coverage data.



Summary & Key Messages

- Phase 1 PoC was a technical validation of the use of federated data (analogy: a “road” can be built within the data “highway”).
- Efforts have shifted to conceptualizing the Pan-Canadian Health Data Highway – key attributes of the Data Highway include the following:
 - Adaptive to various detailed business requirements across the three settings: within PT, across PTs and from PT to PHAC
 - Places policy and business needs first, so that technology is not an impediment to enabling interoperability
 - Accommodates various public health events and scenarios with respect to data access (e.g. emergency data sharing for a public health event)
 - Compatible with other interoperability initiatives such as p-DIR (QR code) and digital wallet
 - Aligns with Infoway's Trust Framework
 - Designed as a robust enterprise-grade platform that meets security and privacy requirements
 - The data highway can extend to many business scenarios beyond public health across GC, such as data sharing related to trade barriers and public safety
 - Allows for incremental adoption within a PT, across PTs and PHAC – i.e., if the decision is to proceed with building the data highway, it can be created with a sub-set of PT stakeholders in stages
 - Minimal business transformation is required for existing PT to PHAC surveillance programs in the near term; allows for business features to be added on over time (e.g. AI capabilities) in order to more proactively respond to public health events as per the Health Minister’s Meeting (HMM) commitment
- Please note that whether or not the Pan-Canadian Health Data Highway will proceed is dependent on the outcomes of the two workstreams depicted earlier in this deck - a report is expected to be compiled for summer 2026.

Annex 1 – Pan-Canadian Health Data Highway Business Assumptions

High-Level Business Assumptions for Pan-Canadian Health Data Highway/"Interim State 1"

- The architecture vision has been guided by the following high-level business assumptions and requirements with respect to the Pan-Canadian Health Data Highway/Interim State 1:
 - A Pan-Canadian "data highway" represents an inter-connection amongst PHAC and at least one immunization registry/repository from each PT, while still retaining PT data sovereignty.
 - This "data highway" enables both PT to PT data transfer and PT to PHAC data access.
 - The focus is on immunization data (beyond MMR and other childhood immunization).
 - The FDArch can access and derive from PT immunization registries/repositories both aggregated data and de-identified record-level data.
 - The FDArch can support the data access for identified row-level data (Protected B health data), e.g. to support PT to PT data access and/or row-level PT to PHAC access.
 - The technical assessment of PT's technology and architecture is out of scope for this FDArch solution architecture assessment.

To create a high-level architecture that supports the above high-level business requirements, the architecture has been designed to be flexible, scalable and extensible to allow for potential future business requirements. For example, the architecture is designed to support access to identifiable record-level data if this becomes a business requirement in the face of a public health emergency. Also, multiple access points within a PT can be onboarded onto the data highway.

Annex 2 – Information about Federated Data Approach

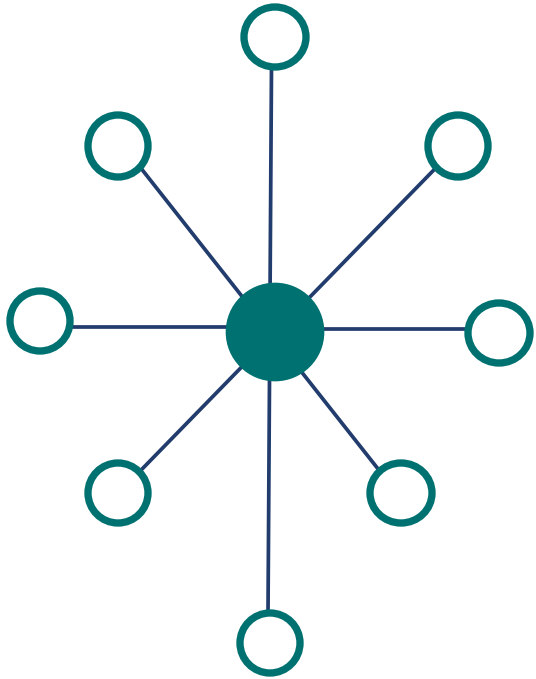
**Solving Our
Data Sharing
Issues**

Public health data is siloed.

Multiple factors contribute to the siloing of public health data, including unclear data ownership, poor system interoperability, and concerns over privacy protection.

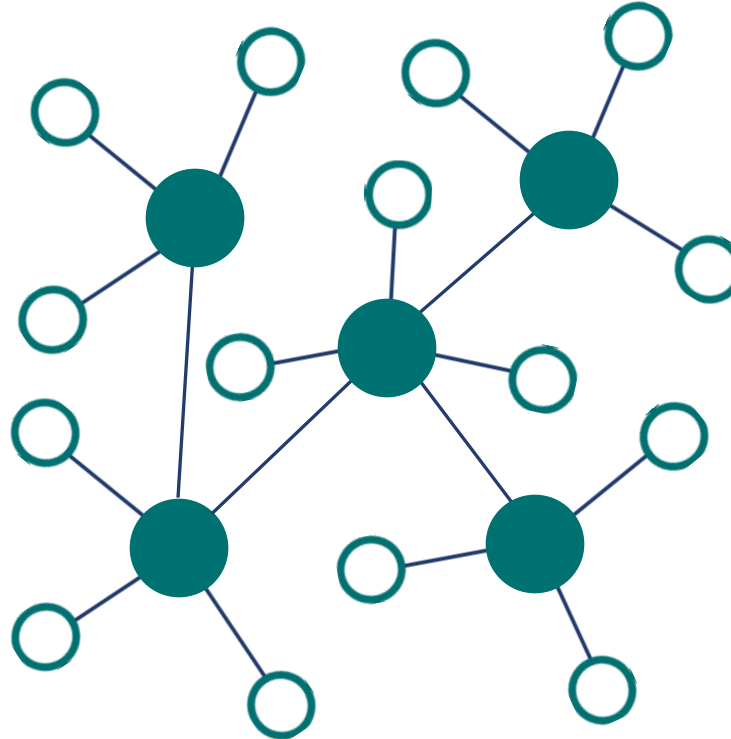
Federated data systems (or networks) are gaining international attention and provide a potential solution to address the siloing of public health data and current barriers to data sharing.

DATA IS GENERALLY SHARED THROUGH THESE IT INFRASTRUCTURE MODELS



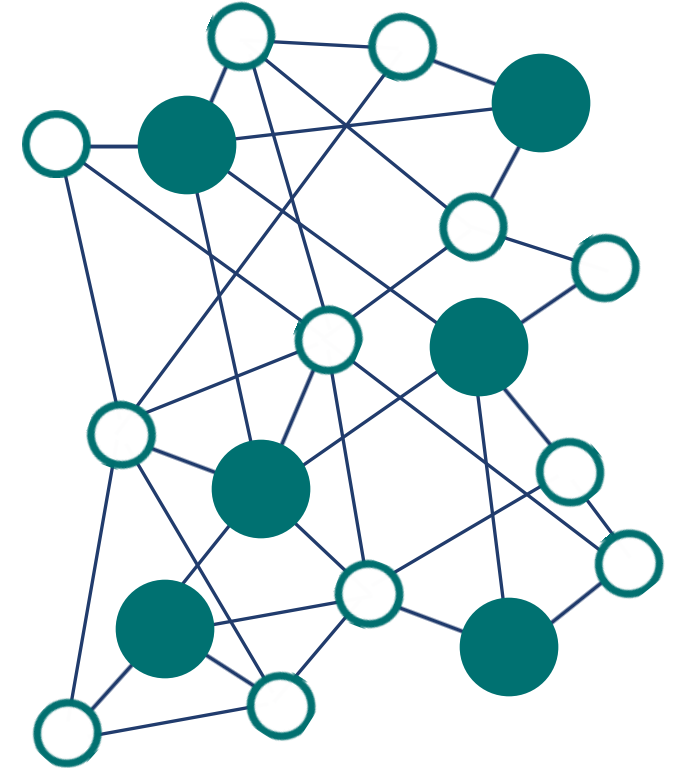
Centralized Model

All the data is stored in a single location and access is managed by a central authority.



Decentralized Model

The data is stored in multiple locations and access is managed by multiple authorities, each one storing a copy of the data.



Federated Model

The data remains at each location but provides access to other users to 'visit' the data, based on agreed-upon privileges.

WHY ADOPT A FEDERATED MODEL FOR DATA SHARING?

Greater Access to Data

Federated networks can facilitate access to public health data for Canadians, health care providers and public health units, enabling large cohort analysis across healthcare institutions, regions, and even national borders, while also promoting data democratization and knowledge sharing.

No One-Size-Fits-All

Federated technical architectures avoids using a “one-size-fits-all” approach to health data platforms to system connectivity via secure, open and reliable means of accessing disparate legacy technology systems and data formats.

Enhanced Privacy and Security

Each organization or provider maintains ownership and control over their data, granting access to users as appropriate. This allows for tailoring to the unique privacy requirements of individual jurisdictions.

Alignment to Division of Powers

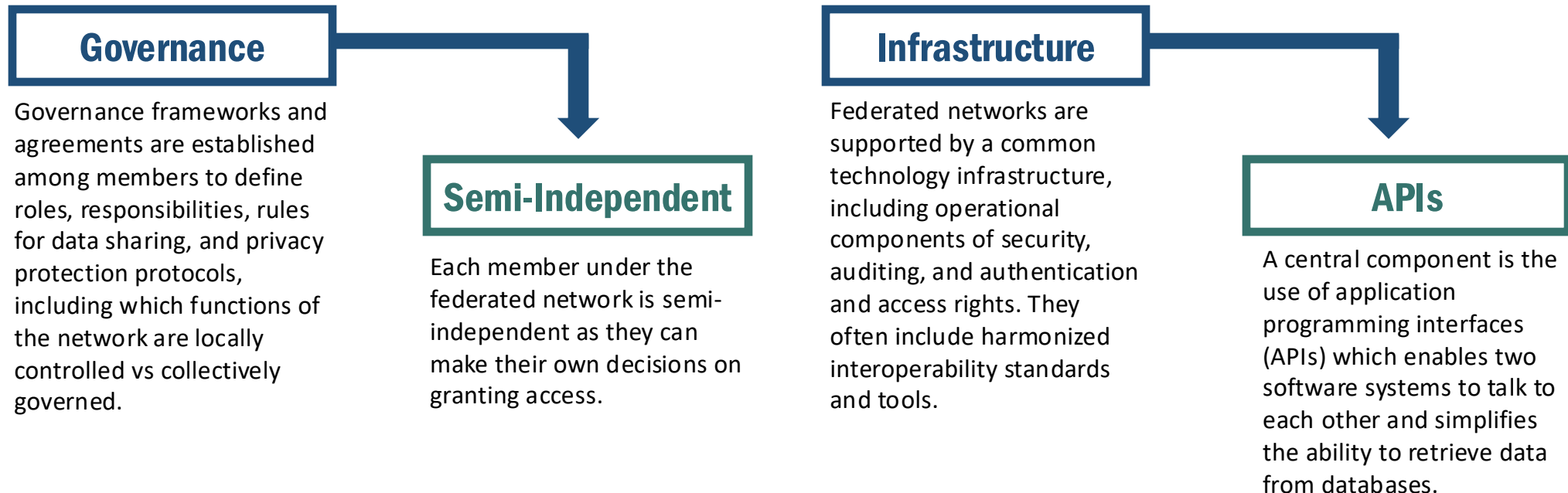
A federated network approach for public health would reflect the shared responsibility for health surveillance across jurisdictions and presents opportunities for sharing infrastructure, without imposing a rigid singular solution.

FEDERATED NETWORKS FACILITATE DATA “VISITING”

A Federated Network is a series of decentralized, interconnected nodes, which allows data to be queried by authorized users in the network without the data leaving its source.

In contrast to traditional data sharing approaches where data is duplicated and shared, a federated network facilitates data access or ‘visiting’ of existing data sets. Each user remains in control over their own data but provides access to other members in the network based on agreed-upon governance frameworks, protocols, and standards.

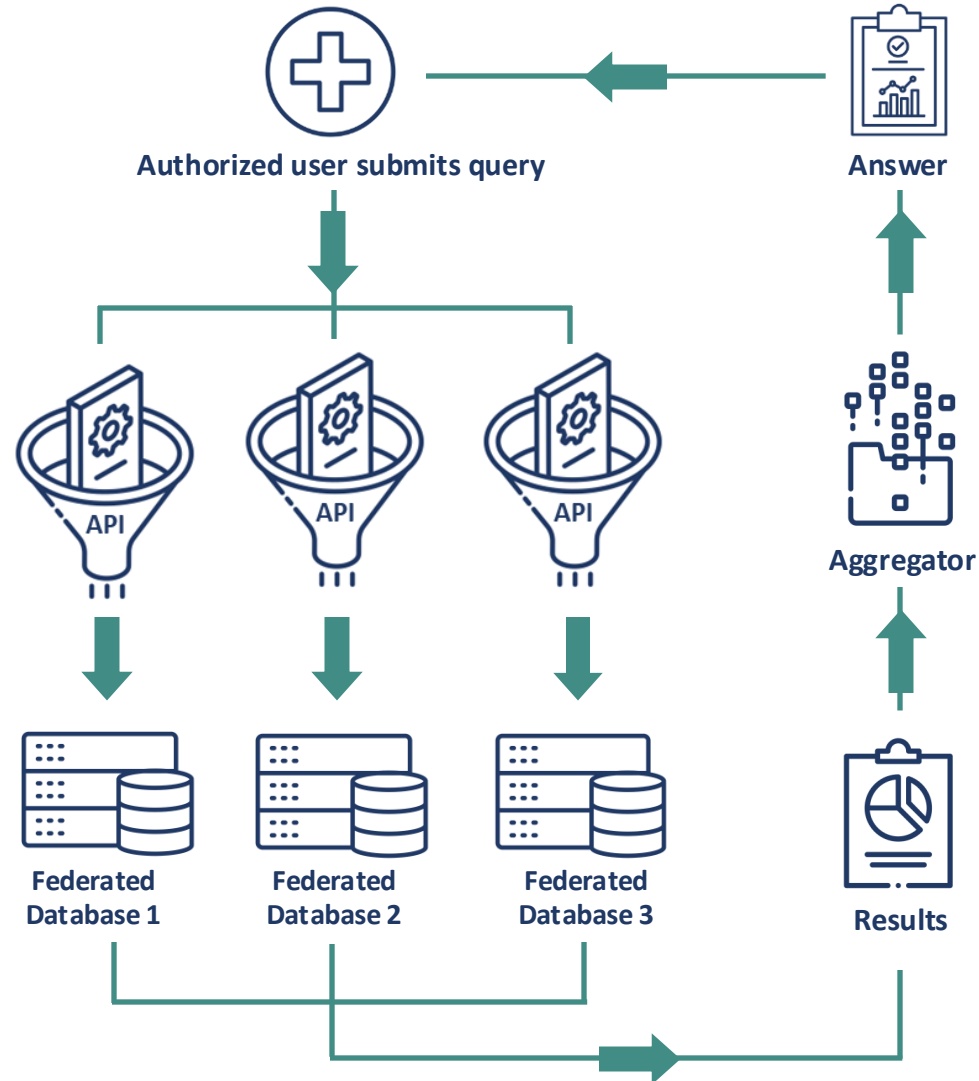
Federated networks generally include the following common characteristics:



BRINGING THE QUESTION TO THE DATA

Authorized users can perform computer/system query for integrated data from all the members within a federated network.

The APIs retrieve the data from many types of databases and applications, including those at remote locations.

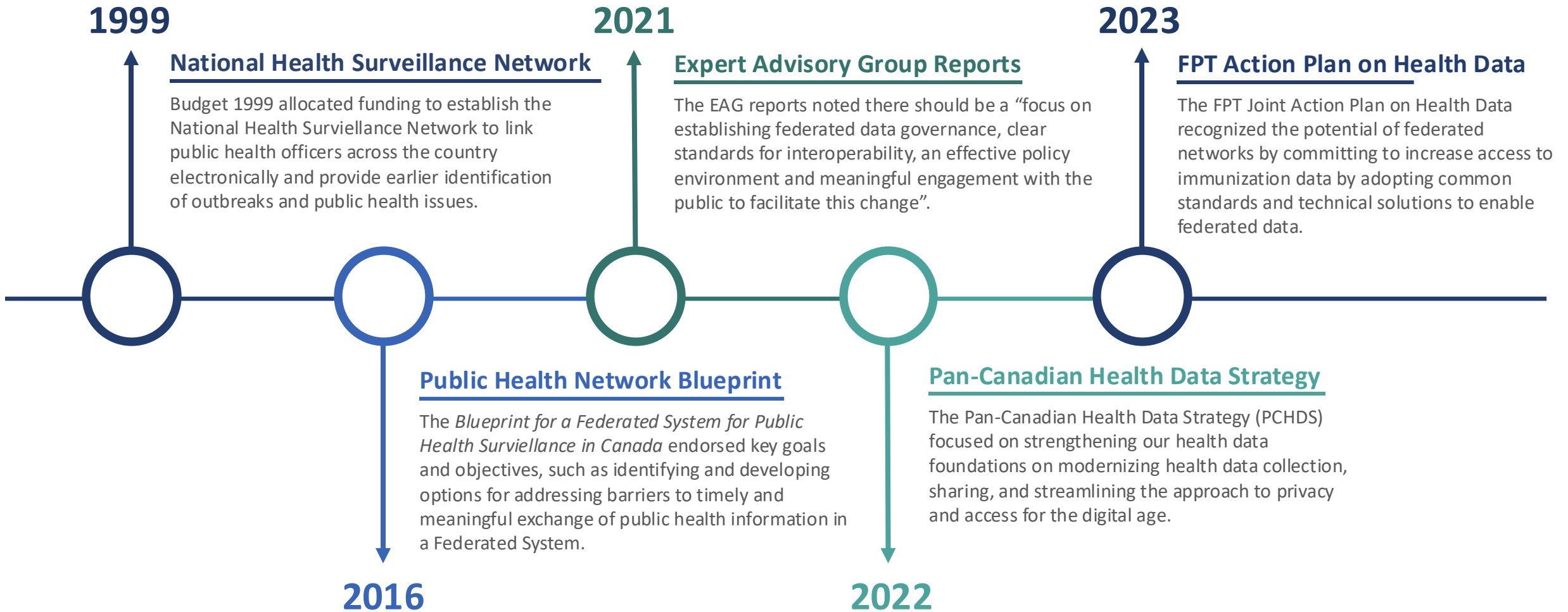


The resulting dataset is returned to the individual who submitted the query often in real-time after which analysis can be applied to answer public health questions.

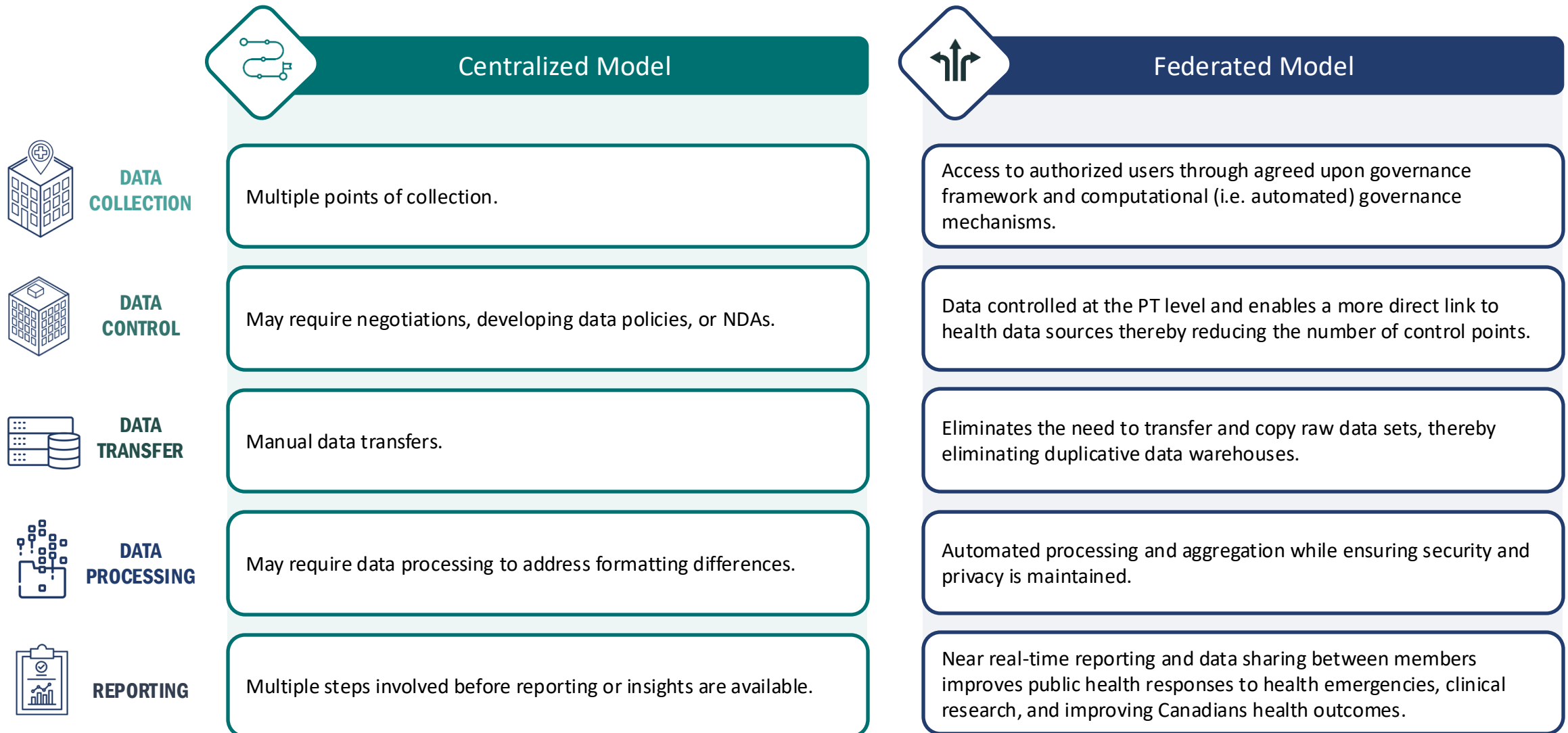
The API can also function as 'middleware' software to aggregate the results.

Participants are only provided with the data they need, based upon agreed needs and requirements of members.

A FEDERATED APPROACH TO CANADA'S HEALTH DATA CONTINUES TO BE ADVOCATED



COMPARING CENTRALIZED VS FEDERATED DATA SHARING MODELS



MOMENTUM TOWARDS FEDERATED NETWORKS IS GROWING



Canada

CanDIG is the first multi-project human genomics and biomedical data federation in Canada, connecting the country's largest human sequencing centers.



United States/International

The Federated Tumor Segmentation initiative is jointly led by Intel and the University of Pennsylvania, with participation from over 20 international collaborating sites. The initiative's goal is to gain knowledge for tumor detection from de-identified personal data.



United Kingdom

The Dare UK Genomics pilot connects two genomics databases in the UK and allows for federated data analysis between them.



Germany

The Kaapana project, working through the Joint Imaging Platform across 36 German university hospitals focuses on radiological and radio therapeutical imaging data analysis.



Europe/Australia

The GenomEUtwin project is a federated network that links eight Genome Twin databases across several European countries and Australia.



RECENT ADVANCEMENTS IN FEDERATED DATA NETWORKS IN HEALTH CARE



“NHS England’s biggest ever IT contract”

In November 2023, the [National Health Service](#) signed a contract with Palantir and its partners for the development of a federated data platform valued at £330M over seven years.

The federated data platform will initially focus on supporting:

1. Elective recovery to address patient backlogs
2. Effective care coordination between local health organizations
3. Supporting the vaccination and immunization for vulnerable people
4. Help integrated care systems proactively plan services that meet the needs of their population.



HDRN launching Federated Analysis Learning Series

Health Data Research Network Canada will be launching a [learning series](#) on federated data in early 2024 with six planned webinars that will provide key insights from experts. The webinars were launched in early January.

The HDRN also developed a federated analysis [resources page](#) that includes current approaches for distributed analysis, statistical review with distributed data, along with links to key trusted research environments and leaders in artificial intelligence with federated data.